



Patrick Dwyer

Mathematician and computer scientist (BA, Northwestern University) working as a data scientist, with a foundation in machine learning, optimization, and scientific computing. Applied experience productionizing ML and NLP research code, building large-scale data pipelines, and developing neural networks in PyTorch — including 3D convolutional networks for biological motion tracking. Increasingly oriented toward research at the intersection of computation and the life sciences, with interests in computational biology and neuroscience. Google Cloud [Certified](#) Professional Machine Learning Engineer.

Education

BA Mathematics, BA Computer Science

Sep 2019 – June 2023

Northwestern University, Evanston, IL — Weinberg College of Arts and Sciences — 3.43/4.00 GPA

Research Experience

Schwartz Lab: Lab Tech

July 2023 – Sep 2023

- Developed a supervised 3D convolutional neural network in PyTorch to track key mouse body parts in 3D from four synchronized camera views
- Extended PyTorch's torch.autograd.Function class to integrate 3D→2D point projection into autograd
- Implemented an extrinsic camera-calibration routine using Levenberg-Marquardt for Bundle Adjustment
- Manually labelled 8,544 ground-truth 2D points with a custom-built image-labeling program

Professional Experience

Manifold Group: Data Scientist

October 2023 – Present

- Designed and implemented internal chat bot timesheet application on GCP & Palantir Foundry, increasing time tracking compliance by 38%
- Productionized NLP research code, designing and architecting a scalable solution that reduced runtime from 6 hours to 30 minutes (~92% improvement), enabling faster iteration and broader production use.
- Implemented vector embedding similarity pipelines from inception to production, unlocking the ability for clients to analyze previously opaque free-form text and derive actionable insights.
- Optimized client stored procedures decreasing query time by 99.8% to enable ingestion of > 100GBs of data
- Led, designed, and implemented enterprise-level data solutions for a Fortune 500 client using PySpark and Microsoft Fabric
- Designed and developed anomaly detection application (Python, Docker, AWS EC2)

Websanity: Web Developer

September 2023 – Present

- Freelance Web Developer for [Rulepop](#), an emerging rules reference platform for tabletop games

Projects

cwmdmcd.patrickdwyer.com

Winter 2025-Present

- Responsive CWD sampling results dashboard built with MDC's public ArcGIS API